Complex Numbers: Complete Solution Sets

Simplifying complex products and powers of the imaginary unit, solving quadratics with complex roots, testing a proposed complex solution, and reporting the complete solution set of a polynomial

Directions.

- Write every complex number in the standard form $a + bi$, with the real part first.
- "Report the complete solution set" means list *every* solution, real and non-real. A degree-n polynomial equation has n solutions counted with multiplicity.
- Remember $i^2 = -1$. That single replacement is what turns an imaginary product back into a real number.
- The problems mix methods on purpose: decide first whether the equation wants square roots, factoring, or the quadratic formula.

1. Multiply and write the result in the form $a + bi$: $(2 + 3i)(4 - i)$

Answer: ________________

2. Simplify i^{23}. Write the answer as a single term.

Answer: ________________

3. Solve and report the complete solution set: $x^2 + 49 = 0$

Answer: __________________

4. Solve and report the complete solution set: $(x - 3)^2 + 16 = 0$

Answer: __________________

5. Show by substitution that $x = 1 + 4i$ is a solution of $x^2 - 2x + 17 = 0$. Simplify all the way to a single number; state what that number has to be for the check to succeed.

Answer: __________________

6. Find every zero of $p(x) = x^4 - 16$, real and non-real, and list the complete zero set.

Answer: ________________

7. Solve and report the complete solution set: $x^2 + 4x + 13 = 0$

Answer: ________________

8. Find the complete zero set of $p(x) = x^3 + 4x$. State how many zeros the degree of the polynomial promises, and check that you have that many.

Answer: ________________

9. Ben substituted $x = -3 - 4i$ into $x^2 + 6x + 25$, got the value 32, and concluded that $-3 - 4i$ is not a solution of $x^2 + 6x + 25 = 0$. Redo the substitution correctly, report the value you get, and state in one sentence what Ben did wrong.

Answer: __________________

10. Find the complete solution set of $x^4 + 3x^2 - 4 = 0$. (Hint: the equation is quadratic in x^2.)

Answer: __________________

11. Solve and report the complete solution set: $4x^2 - 4x + 5 = 0$

Answer: ________________

12. **Synthesis.** A classmate tells you that $x = 3i$ is one zero of $p(x) = x^4 - 2x^3 + 10x^2 - 18x + 9$. Use that fact — and what it forces about a second zero — to find the complete solution set of $p(x) = 0$. Say which solution is repeated.

Answer: ________________